

## CHIANG RAI, Thailand

WMO: 483030

Lat: 19.9614N

Lon: 99.8811E

Elev: 1289

StdP: 14.02

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 1             | 52.1       | 54.8 | 45.8                          | 47.7 | 71.0 | 48.8 | 53.4 | 67.9 | 10.6                  | 69.7 | 9.1 | 76.9 | 0.2                      | 20   | 0.266 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 6             | 12.8                   | 97.4            | 74.0 | 95.3 | 74.0 | 93.2 | 74.4 | 79.7                | 89.0 | 79.0 | 88.1 | 78.5 | 87.2 | 4.2                  | 150  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 77.3                            | 149.2 | 83.5 | 76.8 | 146.5 | 82.9 | 76.3 | 144.2 | 82.4 | 44.4          | 89.2 | 43.8 | 88.6 | 43.2 | 87.5 | 85.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%  | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 9.9               | 8.5  | 7.5 | 48.2                       | 100.6 | 2.6                | 2.2 | 46.4                                               | 102.1 | 44.8       | 103.4 | 43.4       | 104.6 | 41.5       | 106.2 |
|                   |      |     | 47.2                       | 81.5  | 2.6                | 1.2 | 45.3                                               | 82.4  | 43.8       | 83.1  | 42.4       | 83.8  | 40.6       | 84.7  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 78.3   | 70.4  | 73.6  | 78.2  | 82.4  | 82.8  | 82.6  | 81.5  | 81.1  | 81.0  | 79.4  | 75.2  | 70.8  |
|                                                                        | DBStd       | 5.40   | 3.33  | 3.03  | 3.34  | 3.41  | 3.50  | 2.24  | 2.57  | 2.16  | 2.24  | 2.57  | 3.78  | 4.42  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 14     | 5     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 9     |
|                                                                        | CDD50       | 10321  | 634   | 662   | 874   | 973   | 1016  | 978   | 975   | 964   | 931   | 913   | 756   | 645   |
|                                                                        | CDD65       | 4861   | 174   | 242   | 410   | 523   | 551   | 528   | 510   | 499   | 481   | 448   | 307   | 188   |
|                                                                        | CDH74       | 44330  | 1275  | 2187  | 4006  | 5639  | 5685  | 5363  | 4755  | 4438  | 4204  | 3482  | 2105  | 1191  |
|                                                                        | CDH80       | 17157  | 323   | 901   | 2007  | 2965  | 2563  | 1979  | 1514  | 1310  | 1402  | 1212  | 692   | 289   |
| (14) Wind                                                              | WSAvg       | 1.6    | 1.2   | 1.4   | 1.6   | 1.9   | 2.0   | 2.0   | 2.0   | 1.8   | 1.5   | 1.5   | 1.4   | 1.3   |
| (15) Precipitation                                                     | PrecAvg     | 65.7   | 0.5   | 0.6   | 1.5   | 3.5   | 8.7   | 6.9   | 12.1  | 13.7  | 10.9  | 5.0   | 1.8   | 0.7   |
|                                                                        | PrecMax     | 86.5   | 2.4   | 3.7   | 4.5   | 8.2   | 23.8  | 12.9  | 18.0  | 26.1  | 20.0  | 10.0  | 8.5   | 3.8   |
|                                                                        | PrecMin     | 53.1   | 0.0   | 0.0   | 0.0   | 0.1   | 0.9   | 1.2   | 6.0   | 6.1   | 3.8   | 0.7   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 8.9    | 0.7   | 0.9   | 1.3   | 2.0   | 4.6   | 2.9   | 2.8   | 4.3   | 3.8   | 2.3   | 1.8   | 1.1   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 88.0  | 93.1  | 97.4  | 101.0 | 101.3 | 95.8  | 94.3  | 92.0  | 92.2  | 91.5  | 90.6  | 87.8  |
|                                                                        |             | MCWB   | 68.0  | 67.1  | 69.9  | 73.6  | 76.2  | 78.0  | 77.7  | 78.7  | 78.4  | 76.9  | 74.8  | 72.1  |
|                                                                        | 2%          | DB     | 85.5  | 90.7  | 95.2  | 98.2  | 96.6  | 92.7  | 91.4  | 90.0  | 90.5  | 89.7  | 88.2  | 85.5  |
|                                                                        |             | MCWB   | 67.7  | 67.1  | 69.9  | 73.7  | 76.6  | 77.8  | 78.1  | 78.3  | 77.9  | 76.3  | 73.5  | 70.6  |
|                                                                        | 5%          | DB     | 83.7  | 88.6  | 93.1  | 95.9  | 93.7  | 90.6  | 89.3  | 88.2  | 89.0  | 88.3  | 86.4  | 83.4  |
|                                                                        |             | MCWB   | 67.3  | 67.3  | 70.0  | 74.0  | 77.0  | 77.7  | 77.8  | 78.0  | 77.5  | 75.8  | 72.7  | 69.3  |
|                                                                        | 10%         | DB     | 81.6  | 86.3  | 90.7  | 93.5  | 91.1  | 88.7  | 87.3  | 86.4  | 87.3  | 86.6  | 84.4  | 81.0  |
|                                                                        |             | MCWB   | 66.8  | 67.4  | 70.2  | 74.3  | 76.9  | 77.4  | 77.4  | 77.7  | 77.2  | 75.3  | 71.7  | 68.1  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 72.9  | 74.1  | 75.9  | 79.1  | 80.0  | 80.1  | 80.2  | 80.1  | 80.0  | 78.9  | 76.6  | 74.8  |
|                                                                        |             | MCDB   | 81.4  | 85.8  | 89.4  | 92.0  | 92.1  | 90.0  | 88.9  | 88.3  | 88.5  | 87.9  | 86.1  | 84.2  |
|                                                                        | 2%          | WB     | 70.5  | 72.0  | 74.4  | 77.6  | 79.0  | 79.1  | 79.1  | 79.2  | 79.1  | 77.8  | 75.6  | 72.8  |
|                                                                        |             | MCDB   | 80.3  | 83.5  | 87.6  | 90.9  | 90.8  | 88.6  | 87.5  | 86.9  | 87.5  | 86.5  | 84.5  | 81.2  |
|                                                                        | 5%          | WB     | 69.2  | 70.3  | 73.3  | 76.5  | 78.1  | 78.4  | 78.4  | 78.6  | 78.4  | 77.1  | 74.7  | 71.3  |
|                                                                        |             | MCDB   | 79.4  | 81.8  | 86.4  | 89.9  | 89.2  | 87.2  | 86.2  | 86.0  | 86.3  | 85.5  | 82.9  | 79.6  |
|                                                                        | 10%         | WB     | 68.0  | 68.9  | 72.0  | 75.7  | 77.4  | 77.9  | 77.9  | 78.0  | 77.8  | 76.4  | 73.6  | 69.9  |
|                                                                        |             | MCDB   | 78.0  | 80.9  | 85.3  | 88.7  | 87.5  | 86.1  | 85.0  | 84.7  | 85.1  | 83.9  | 80.9  | 77.8  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 25.0  | 28.3  | 26.7  | 22.3  | 16.2  | 12.8  | 11.1  | 11.1  | 13.0  | 15.2  | 19.4  | 21.8  |
|                                                                        |             | MCDBR  | 27.9  | 31.4  | 30.6  | 25.9  | 20.7  | 15.7  | 14.6  | 13.8  | 15.4  | 17.3  | 21.3  | 23.7  |
|                                                                        |             | MCWBR  | 11.0  | 10.7  | 9.3   | 7.0   | 6.0   | 4.5   | 4.5   | 4.6   | 5.1   | 5.5   | 7.6   | 9.3   |
|                                                                        | 5% WB       | MCDBR  | 22.5  | 25.7  | 24.4  | 21.8  | 17.3  | 13.6  | 12.7  | 12.5  | 14.1  | 15.1  | 17.5  | 19.8  |
|                                                                        |             | MCWBR  | 9.5   | 10.0  | 8.8   | 7.6   | 5.9   | 4.6   | 4.5   | 4.7   | 5.1   | 5.4   | 6.5   | 8.0   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.420 | 0.502 | 0.705 | 0.672 | 0.491 | 0.439 | 0.433 | 0.456 | 0.473 | 0.495 | 0.418 | 0.413 |
|                                                                        | taud        |        | 2.233 | 2.002 | 1.557 | 1.681 | 2.205 | 2.384 | 2.389 | 2.305 | 2.232 | 2.130 | 2.303 | 2.277 |
|                                                                        | Ebn at Noon |        | 270   | 252   | 208   | 218   | 259   | 271   | 272   | 268   | 263   | 252   | 268   | 267   |
|                                                                        | Edh at Noon |        | 42    | 55    | 89    | 80    | 47    | 39    | 38    | 42    | 45    | 48    | 38    | 39    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1415  | 1692  | 1745  | 1886  | 1707  | 1496  | 1354  | 1361  | 1434  | 1397  | 1417  | 1310  |
|                                                                        | RadStd      |        | 78    | 83    | 98    | 117   | 153   | 105   | 104   | 87    | 64    | 66    | 105   | 91    |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | +0.58 | +1.40   | N/A    | N/A     | +0.32 | +0.25 | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (10 neighbors) |  | +0.59 | +1.13   | N/A    | N/A     | +0.22 | +0.34 | N/A         | N/A   | +284  | +281  |

Nomenclature: See separate page